

MDI3003

ADVANCED PREDICTIVE ANALYTICS

LABORATORY ASSESSMENT – EXPERIMENT 05

Tweet Sentiment Analysis

Verified Report Based on the Completed Notebook

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GitHub: https://github.com/Shobana0212/lab5_Tweet-Sentiment-Analysis.git

Abstract

This study presents a reproducible Natural Language Processing (NLP) approach for **three-class sentiment classification of tweets** using the **Twitter US Airline Sentiment dataset**. The objective is to automatically determine whether a tweet expresses **negative, neutral, or positive sentiment** toward an airline service entity. The completed experimental workflow contains **14,640 tweets** and follows a leakage-safe methodology in which the data is audited, minimally normalized, and divided using a fixed **80:20 stratified train-test split**. The training portion is further evaluated using **five-fold Stratified Cross-Validation**, while the test set containing **2,928 tweets** is kept locked until final evaluation. This ensures that model comparison and selection are not influenced by the final test results.

The study compares multiple approaches representing different levels of complexity. **DummyClassifier** is used as a trivial baseline, while **VADER** provides a lexicon- and rule-based social-media sentiment baseline without model training. Three classical machine-learning approaches are then evaluated using **TF-IDF representations: Multinomial Naive Bayes, Logistic Regression, and LinearSVC**. The TF-IDF transformation is incorporated directly inside the model pipelines so that vocabulary and statistical information are learned only from the appropriate training folds, preventing data leakage. Model selection is based primarily on **Macro F1**, which gives equal importance to negative, neutral, and positive classes and is therefore more informative than accuracy for the imbalanced dataset.

Among the learned classical models, **Logistic Regression** achieved the highest five-fold cross-validation Macro F1 of **0.744384 ± 0.008241** and was therefore selected as the final classical model. When evaluated on the previously locked test set of **2,928 tweets**, it achieved an **accuracy of 0.7917, Macro F1 of 0.7442, and Weighted F1 of 0.7958**. The class-wise analysis shows that the model performs particularly well for **negative sentiment**, obtaining an F1-score of **0.8655**. In comparison, **positive sentiment** obtained an F1-score of **0.7199**, while **neutral sentiment** was the most difficult class with an F1-score of **0.6472**. This difference demonstrates why class-wise metrics and Macro F1 are necessary in addition to overall accuracy.

Beyond aggregate performance, the experiment includes several forms of error and robustness analysis. A controlled **hashtag/emoji ablation experiment** evaluates whether removing common social-media signals affects performance. Entity-level analysis examines classification behavior separately for the different airlines represented in the dataset. Representative incorrect predictions are also inspected to identify recurring sources of model failure. In addition, a linguistic challenge analysis considers tweets containing **negation, sarcasm hints, emojis, hashtags, product or brand handles, elongated words, and mixed positive-negative sentiment cues**. These analyses provide a more detailed understanding of where the classifier succeeds and where further modeling improvements may be required.

An additional **BiLSTM deep-learning extension** was completed to compare a sequence-based neural architecture with the classical sparse-text models. The BiLSTM achieved a **Macro F1 of 0.7378, Weighted F1 of 0.7910, and Accuracy of 0.7883** on the test set. Thus, in the completed experiment, the BiLSTM did not outperform the selected Logistic Regression model, showing that increased model complexity does not automatically guarantee better performance for this dataset.

Finally, an advanced **Twitter-domain Transformer extension using vinai/bertweet-base** was prepared. The dataset was tokenized using the corresponding pretrained tokenizer, with a maximum sequence length of 128 tokens, and the training/validation/test structure was prepared without altering the locked test set. However, Transformer training did not successfully execute because TrainingArguments was not defined in the executed

notebook state. Consequently, **no Transformer accuracy, Macro F1, or Weighted F1 is reported**, avoiding unsupported or fabricated performance claims. Overall, the experiment demonstrates that a carefully designed classical NLP pipeline can provide strong and interpretable sentiment classification performance while maintaining reproducibility, leakage control, and meaningful error analysis.

1. Introduction

Twitter is an important source of customer feedback because users frequently express their opinions about products, brands, and services through short messages. **Tweet sentiment analysis** is therefore formulated as a supervised classification problem in which the sentiment expressed in a tweet is classified as **negative, neutral, or positive**. This experiment focuses on the **Twitter US Airline Sentiment dataset**, which is particularly suitable because each tweet is associated with an identifiable airline service entity. The laboratory specification defines the task as predicting sentiment toward a named product, brand, service, or entity using reproducible NLP pipelines.

Tweets are different from conventional text because they are short, informal, and often contain **hashtags, mentions, URLs, emojis, abbreviations, repeated characters, punctuation, negation, and sarcasm**. These characteristics can make sentiment classification difficult, as individual words may not always represent the sentiment of the complete message. Therefore, this experiment uses minimal text normalization while preserving meaningful sentiment cues rather than applying aggressive preprocessing.

A major focus of the study is **data quality and leakage prevention**. Before model development, the dataset is audited for missing values, duplicate records, class distribution, and potentially inappropriate fields. Identifiers such as `tweet_id` and target-related annotation information are excluded from predictors to prevent memorization or target leakage. The laboratory specification specifically requires identifiers and leakage-prone annotation fields to be excluded from model inputs.

The experimental workflow uses a fixed **80:20 stratified train-test split**, with the test set kept locked until final evaluation. The training data is evaluated using **five-fold Stratified Cross-Validation**, ensuring that all learned models are compared using the same validation structure. TF-IDF vectorization is performed inside machine-learning pipelines so that vocabulary and statistical information are learned only from the appropriate training data. This provides a leakage-safe and reproducible evaluation procedure.

The study first establishes **DummyClassifier and VADER** baselines and then compares classical TF-IDF-based models, including **Multinomial Naive Bayes, Logistic Regression, and LinearSVC**. Model selection is based primarily on **Macro F1**, because the three sentiment classes are not equally represented in the dataset. Accuracy, Weighted F1, class-wise precision and recall, and confusion matrices are also used to provide a more complete evaluation. The laboratory requires comparison of Logistic Regression and LinearSVC on the same split and recommends MultinomialNB as an additional probabilistic comparison.

In addition to overall performance, the experiment performs **error and entity-level analysis**. Difficult linguistic patterns such as negation, sarcasm, emojis, hashtags, elongated words, and mixed sentiment are examined. Since airline information is available, model performance is also analyzed across individual airlines. These analyses help identify where the classifier performs well and where improvements may be required.

An advanced extension compares the classical approach with a **BiLSTM** sequence model and prepares a **BERTweet Transformer** model. This follows the laboratory's recommended progression from transparent sparse-text baselines to more complex neural and Transformer representations.

Overall, the experiment aims to develop a **reproducible, leakage-safe, and interpretable sentiment-classification workflow** that can support aggregate analysis of airline-related customer opinions. The results should be interpreted carefully because social-media data may contain sampling bias, annotation noise, sarcasm, and domain shift, and therefore does not necessarily represent the opinions of all customers.

2. Objectives

- Formulate airline-service sentiment as a three-class supervised classification task.
- Audit the dataset for nulls, duplicate IDs, duplicate text and leakage/identifier fields.
- Apply minimal tweet normalization while retaining sentiment-bearing information.
- Compare DummyClassifier and VADER baselines with TF-IDF-based learned models.
- Use identical five-fold stratified CV folds for all learned models.
- Select the best classical model using CV Macro F1 rather than final-test performance.
- Evaluate the selected model once on the locked test set.
- Perform per-class, confusion-matrix, error, entity, challenge and confidence analyses.
- Evaluate a controlled hashtag/emoji preprocessing ablation.
- Document BiLSTM and Transformer extensions without reporting unfinished results as completed results.

3. Dataset Description and Provenance

The **Twitter US Airline Sentiment** dataset is used as the core dataset for this experiment because it directly represents sentiment expressed by customers toward identifiable airline service entities. This makes it suitable for the laboratory objective of predicting whether a tweet expresses **negative, neutral, or positive sentiment** toward a product, brand, service, or named entity. The laboratory manual identifies Twitter US Airline Sentiment as the preferred core dataset for the three-hour experiment because it provides direct business/service-entity context.

The dataset consists of customer tweets related to six U.S. airlines and was collected during **February 2015**. The notebook successfully loaded **14,640 observations** and retained the fields `tweet_id`, `text`, `airline_sentiment`, and `airline` for the working dataset. Here, `text` represents the tweet content used for prediction, `airline_sentiment` is the target variable, and `airline` provides the service/entity context required for entity-level analysis.

The notebook records the source as an **instructor-approved public dataset / Kaggle source** and records the access date as **23 August 2026**. However, the exact downloaded dataset version and license/usage terms were not entered into the notebook. Therefore, these details are not assumed or inferred in this report. Recording dataset source, access date, version, labels, and license/usage terms is a required part of the laboratory's dataset governance checklist.

Table 3.1 Dataset Provenance and Characteristics

Attribute	Verified notebook value
Dataset	Twitter US Airline Sentiment
Rows loaded	14,640
Columns retained for working dataset	tweet_id, text, airline_sentiment, airline
Target	airline_sentiment
Entity	airline
Classes	negative, neutral, positive
Collection period	February 2015
Random seed	42

The dataset is therefore suitable for a **three-class supervised classification problem**. Each observation represents a tweet, while the corresponding sentiment label provides the target that the machine-learning models attempt to predict. The airline field additionally makes it possible to investigate whether model performance and predicted sentiment distributions differ across individual service entities.

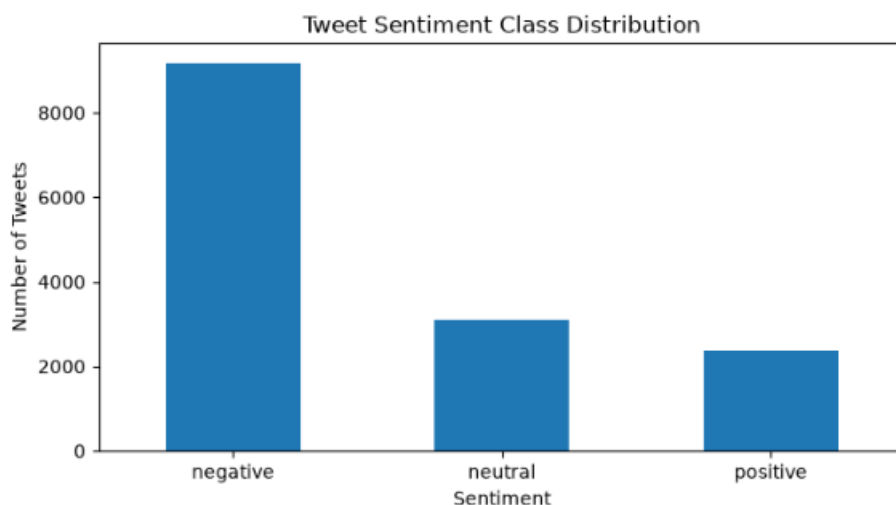
3.1 Class Distribution

An important characteristic of the dataset is its **imbalanced sentiment distribution**. Negative tweets form the majority of the dataset, while positive tweets represent the smallest class. The complete dataset contains **9,178 negative tweets, 3,099 neutral tweets, and 2,363 positive tweets**.

Table 3.2 Class Distribution

Class	Count	Percentage
Negative	9,178	62.69%
Neutral	3,099	21.17%
Positive	2,363	16.14%

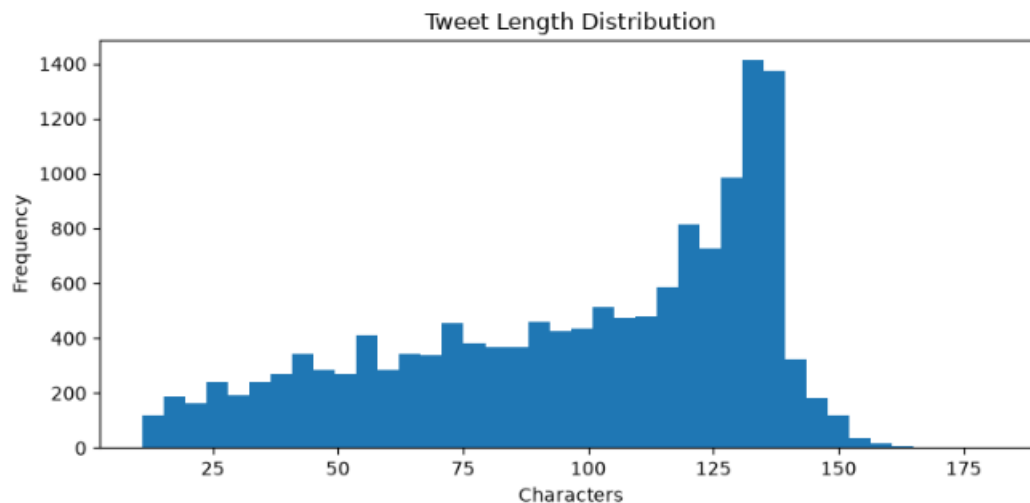
Approximately **62.69%** of the observations are negative, meaning that a classifier could obtain relatively high overall accuracy simply by favoring the majority class. Such a result would not necessarily indicate good performance on neutral and positive sentiment. Therefore, relying only on accuracy would provide an incomplete evaluation of the classifier.



To address this imbalance, the experiment uses **Macro F1 as the primary model-selection metric**. Macro F1 calculates the F1-score independently for each sentiment class and then gives each class equal importance. This makes it particularly useful for determining whether the classifier performs consistently across negative, neutral, and positive tweets. **Weighted F1, accuracy, precision, recall, and confusion matrices** are also reported to provide complementary information.

The imbalance also motivates the use of **class weighting** in the Logistic Regression and LinearSVC models. This allows the learned classifiers to give greater consideration to the less frequent sentiment categories instead of allowing the large negative class to dominate the learning process.

The dataset's imbalance is also important when interpreting the final business analysis. A high proportion of negative tweets in this dataset should **not** be interpreted as evidence that 62.69% of all airline customers are dissatisfied. The laboratory manual explicitly notes that social-media samples are not necessarily representative of all customers and that observed sentiment should not be treated as a complete measure of customer satisfaction.



Thus, the dataset provides an appropriate and well-defined benchmark for evaluating tweet sentiment classifiers, while its class imbalance, historical collection period, and social-media sampling characteristics must be considered when interpreting the experimental results.

4. Data Audit and Leakage Assessment

Before developing the sentiment-classification models, a systematic audit was performed to assess the **quality, completeness, uniqueness, and suitability of the dataset for machine-learning prediction**. The audit is an important stage because errors in the input data or the inclusion of information derived from the target variable can result in misleadingly high model performance.

The notebook loaded **14,640 observations across four working columns**: `tweet_id`, `text`, `airline_sentiment`, and `airline`. The audit confirmed that there were **no missing values** in any of these four fields. Therefore, the core dataset did not require missing-value imputation for the variables used in the experiment.

Table 4.1 Data Audit Results

Audit item	Verified result
Dataset shape	14,640 × 4 working columns
Null tweet_id	0
Null text	0
Null sentiment	0
Null airline	0
Duplicate tweet IDs	155
Duplicate raw text	213
Duplicate cleaned text	336
Potential excluded field found	tweet_id
Other listed leakage fields present	None

4.1 Missing-Value Assessment

The audit found **zero null values** for tweet_id, tweet text, sentiment, and airline. This indicates that every observation used in the working dataset has the basic information required for the classification task.

In particular, there are no tweets with missing sentiment labels or missing airline information. Therefore, the notebook does not need to perform deletion or imputation based on missing target values before constructing the classification models.

4.2 Duplicate Record Assessment

The audit identified **155 duplicate tweet IDs**, **213 duplicate raw-text records**, and **336 duplicate cleaned-text records**. The increase in duplicate cleaned text compared with raw text indicates that normalization can cause different original strings to become identical after preprocessing.

These duplicate findings are recorded as part of the dataset audit. Importantly, the notebook does **not explicitly state that these duplicate observations were removed from the core experiment**. Therefore, this report does not claim that duplicate records were deleted or deduplicated before model training.

This distinction is important for reproducibility. Reporting the audit result while clearly documenting whether records were actually removed prevents the report from implying a preprocessing step that was not performed.

4.3 Identifier and Leakage Assessment

The field tweet_id is an identifier rather than a linguistic feature. It was therefore **excluded from the predictors**. Allowing an identifier to enter the model could permit memorization of individual records without providing meaningful information about the sentiment expressed in the tweet.

The notebook audit did not identify other listed leakage-prone fields in the working dataset. Fields such as annotation confidence, negative-reason metadata, usernames, coordinates, and other target-derived information should not be used as sentiment predictors because they either reveal information about the target or introduce unnecessary privacy and fairness concerns.

The laboratory specification explicitly requires identifiers and leakage/annotation fields to be removed from predictors and requires the TF-IDF transformation to be fitted only within the training/CV pipeline.

4.4 Leakage-Safe Experimental Design

The experiment also prevents **train-test information leakage** through its evaluation design. The complete dataset is divided into training and test portions before final model evaluation. The test set is treated as a **locked set** and is not used for model selection.

For the learned classical models, TF-IDF vectorization is contained within the corresponding scikit-learn Pipeline. Consequently, vocabulary construction and TF-IDF statistics are learned independently within each training fold during cross-validation rather than being calculated using the entire dataset beforehand.

This is consistent with the methodology emphasized in the reference report, where predictors must not contain post-outcome information and model development must maintain a strict separation between training and testing data.

4.5 Audit Summary

Overall, the dataset audit shows that the working dataset is **complete with respect to the four required fields**, but contains identifiable duplicate IDs and text records that must be documented. The core experiment retains these observations because the notebook does not document their removal. The tweet_id identifier is excluded from prediction, and no other listed leakage fields are present in the working dataset.

Therefore, the subsequent modeling stage is based on a clearly documented dataset with **14,640 observations**, while maintaining the required separation between identifiers, predictors, target information, and the locked evaluation data.

5. Tweet Preprocessing

Tweet preprocessing is an important stage in the sentiment-analysis pipeline because social-media text contains considerable formatting and linguistic variation. However, the preprocessing used in this experiment is intentionally **minimal**. The objective is to reduce irrelevant textual variation while preserving information that may contribute to sentiment classification. The laboratory specification specifically recommends preserving negation, emojis, hashtags, punctuation, and product handles unless an ablation study provides evidence for removing them.

The preprocessing applied to the tweet text consists primarily of normalization of URLs, mentions, and whitespace. URLs are replaced with the special token <URL>, while user mentions are replaced with <USER>. This prevents the classifier from memorizing individual URLs or user identifiers while retaining the information that a URL or user mention occurred in the tweet. Repeated or unnecessary whitespace is normalized to produce a more consistent text representation.

Table 5.1 Preprocessing Operations

Operation	Verified treatment
URLs	Replace with <URL>
Mentions	Replace with <USER>
Whitespace	Normalize
Negation	Preserved
Emojis	Preserved in the core representation
Hashtags	Preserved in the core representation

Punctuation	Preserved
Product/brand handles	Preserved through normalized mention placeholder

The normalized text is used by the core classifiers. The notebook reports 336 duplicate cleaned-text records after normalization.

6. Train/Test and Cross-Validation Strategy

A reliable sentiment-classification experiment requires a clear separation between **model development** and **final model evaluation**. In this study, the complete Twitter US Airline Sentiment dataset containing **14,640 tweets** is divided into training and testing subsets using a fixed **80:20 stratified split**. Stratification ensures that the proportions of negative, neutral, and positive tweets are maintained as closely as possible in both subsets.

A fixed random state of **42** is used to make the split reproducible. The resulting training set contains **11,712 tweets**, while the locked test set contains **2,928 tweets**. The test set is not used during model selection and is reserved for the final evaluation of the selected model.

Table 6.1 Train/Test Configuration

Parameter	Verified value
Total observations	14,640
Training observations	11,712
Locked test observations	2,928
Split	80:20
Sampling	Stratified
Random state	42
Training class counts	Negative 7,343; Neutral 2,479; Positive 1,890
Cross-validation	5-fold StratifiedKfold
Test usage	Final evaluation after model selection

The training data contains 7,343 negative, 2,479 neutral, and 1,890 positive tweets. The stratified procedure ensures that all three sentiment classes are represented in the training and test partitions according to their original distribution.

Table 6.2 Training Class Distribution

Sentiment	Training Count
Negative	7,343
Neutral	2,479
Positive	1,890
Total	11,712

After creating the fixed training set, five-fold Stratified Cross-Validation is used to compare the learned models. In each fold, four portions of the training data are used for learning and the remaining portion is used for validation. The process is repeated five times so that every training observation participates in validation once.

Importantly, the locked test set is not used during cross-validation or model selection. This prevents the final test performance from influencing the choice of model. The approach is consistent with the reference report's methodology, which emphasizes maintaining a strict separation between training and testing data and using stratified cross-validation for reliable model development.

The same cross-validation folds are used for Logistic Regression, LinearSVC, and MultinomialNB, allowing their performance and stability to be compared fairly. TF-IDF is fitted inside each model's pipeline, ensuring that vocabulary and feature statistics are learned only from the corresponding training portion of each fold.

7. Baseline Models

Before evaluating the learned machine-learning classifiers, two baseline approaches are established: **DummyClassifier** and **VADER**. These baselines provide reference points against which the more sophisticated models can be evaluated.

The **DummyClassifier** provides a trivial classification strategy and establishes the performance that can be obtained without learning meaningful linguistic relationships. It achieved a test accuracy of **0.6267**, Macro F1 of **0.2568**, and Weighted F1 of **0.4829**.

VADER is a lexicon- and rule-based sentiment analyzer designed for social-media text. It does not require supervised training on the airline dataset. It achieved an accuracy of **0.5523**, Macro F1 of **0.5206**, and Weighted F1 of **0.5730**.

Table 7.1 Baseline Performance

Model	Test Accuracy	Test Macro F1	Test Weighted F1
DummyClassifier	0.6267	0.2568	0.4829
VADER	0.5523	0.5206	0.5730

Although the DummyClassifier has higher accuracy than VADER, its very low Macro F1 demonstrates that it performs poorly when equal importance is given to all three sentiment classes. VADER achieves a considerably higher Macro F1, indicating better balance across the classes, but it still remains below the supervised TF-IDF models.

These results demonstrate the value of training classifiers specifically on the airline-domain dataset rather than relying only on a generic sentiment lexicon.

8. Classical TF-IDF Models

The next stage evaluates three supervised classical machine-learning models using **TF-IDF word features**:

1. **Multinomial Naive Bayes**
2. **Logistic Regression**
3. **LinearSVC**

TF-IDF uses **word n-grams of size 1–2**, allowing the models to learn both individual words and two-word expressions. The configuration uses:

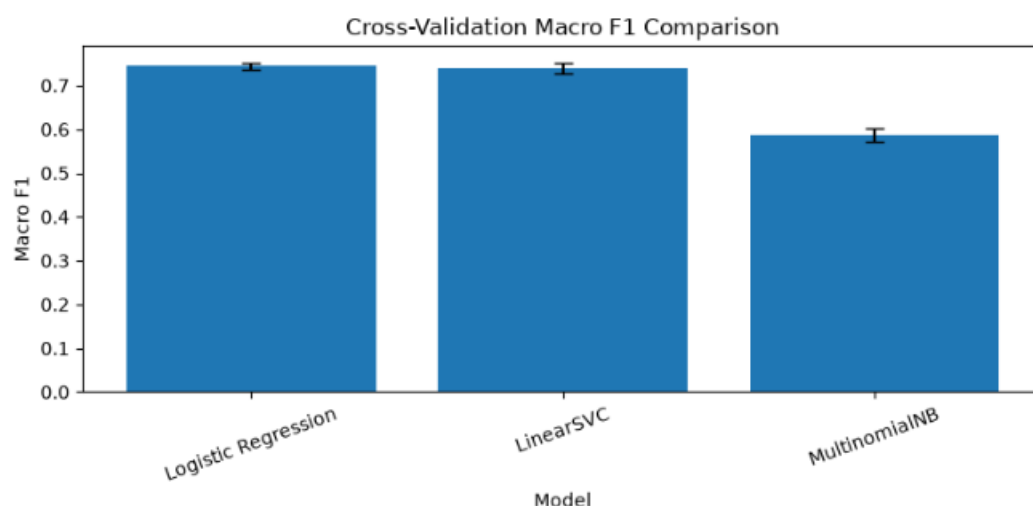
- ngram_range=(1,2)
- min_df=2
- max_df=0.98
- sublinear_tf=True

Logistic Regression and LinearSVC use class_weight='balanced' to reduce the effect of the imbalanced sentiment distribution, with max_iter=2000. MultinomialNB uses alpha=0.5.

Table 8.1 Five-Fold Cross-Validation Results

Model	CV Macro F1	CV SD	CV Weighted F1	CV Accuracy	Mean Fit Time (s)	Wall Time (s)
Logistic Regression	0.744384	0.008241	0.795471	0.792776	3.601085	16.639406
LinearSVC	0.739502	0.011703	0.796877	0.800631	2.702077	3.623467
MultinomialNB	0.585816	0.015062	0.690117	0.736339	2.125534	17.614001

The results show that **Logistic Regression and LinearSVC substantially outperform MultinomialNB in Macro F1**. Logistic Regression obtains the highest Macro F1, while LinearSVC obtains slightly higher CV Accuracy and Weighted F1.



MultinomialNB has the lowest Macro F1 at **0.585816**, indicating weaker balanced performance across the three sentiment classes despite its relatively fast individual fold fitting.

LinearSVC is computationally efficient in this experiment, with a wall time of **3.623467 seconds**, compared with **16.639406 seconds** for Logistic Regression. However, computational speed alone is not the model-selection criterion.

8.1 Model Selection

The laboratory specifies **Macro F1 as the primary metric for selecting the core model** because each sentiment class should receive equal importance. Based on this criterion, **Logistic Regression is selected**.

Logistic Regression achieves a five-fold CV Macro F1 of: **0.744384 ± 0.008241**

LinearSVC achieves: **0.739502 ± 0.011703**

Therefore, Logistic Regression exceeds LinearSVC by approximately **0.004882 Macro F1**.

Although LinearSVC produces a slightly higher CV Accuracy (**0.800631**) and Weighted F1 (**0.796877**), these metrics are secondary to Macro F1 for the specified three-class task. Consequently, selecting Logistic Regression is methodologically correct.

The selected Logistic Regression pipeline is subsequently retrained using the complete training partition and evaluated **once on the locked 2,928-tweet test set**. This preserves the distinction between model selection and final performance estimation.

The use of Macro F1 as the primary selection criterion also addresses the dataset's substantial class imbalance, ensuring that the model is not selected merely because it performs well on the majority negative class. The laboratory requirements specifically identify Macro F1, Weighted F1, per-class metrics, and confusion matrices as required evaluation outputs.

9. Final Locked-Test Evaluation

After completing model selection using the training data and five-fold cross-validation, **Logistic Regression** was selected as the final classical model because it achieved the highest cross-validation Macro F1. The selected pipeline was then evaluated once on the **locked test set containing 2,928 tweets**. This ensures that the reported test performance represents an unbiased final evaluation rather than a value used during model selection.

Table 9.1 Final Test Performance

Metric	Verified value
Selected model	Logistic Regression
Accuracy	0.7917
Macro F1	0.7442
Weighted F1	0.7958
Test size	2,928

9.1 Per-Class Performance

Class	Precision	Recall	F1	Support
Negative	0.890490	0.841962	0.865546	1,835
Neutral	0.590426	0.716129	0.647230	620
Positive	0.746032	0.695560	0.719912	473
Macro average	0.7423	0.7512	0.7442	2,928
Weighted average	0.8036	0.7917	0.7958	2,928

The results show a clear difference in performance across the sentiment classes. **Negative sentiment is classified most effectively**, with an F1-score of **0.865546**. Its high precision of **0.890490** indicates that when the model predicts a tweet as negative, the prediction is usually correct.

The **neutral class is the most difficult class**, with an F1-score of **0.647230**. Although its recall is relatively high at **0.716129**, its precision is only **0.590426**, indicating that a considerable number of tweets predicted as neutral actually belong to another sentiment class.

Positive sentiment obtains an F1-score of **0.719912**, with a precision of **0.746032** and recall of **0.695560**. Thus, the model performs reasonably well for positive tweets but does not identify them as consistently as negative tweets.

The difference between the Weighted F1 (**0.7958**) and Macro F1 (**0.7442**) is also important. Weighted F1 is influenced by the much larger negative class, whereas Macro F1 gives equal importance to all three classes. Therefore, Macro F1 provides a more informative assessment of balanced sentiment-classification performance.

10. Confusion Matrix and Error Analysis

To understand the types of mistakes made by the selected Logistic Regression model, the notebook generates both a **confusion matrix using raw counts** and a **row-normalized confusion matrix**. These matrices provide complementary information: the count matrix shows the number of predictions in each category, while the normalized matrix makes it easier to compare classification behavior across classes with different support.

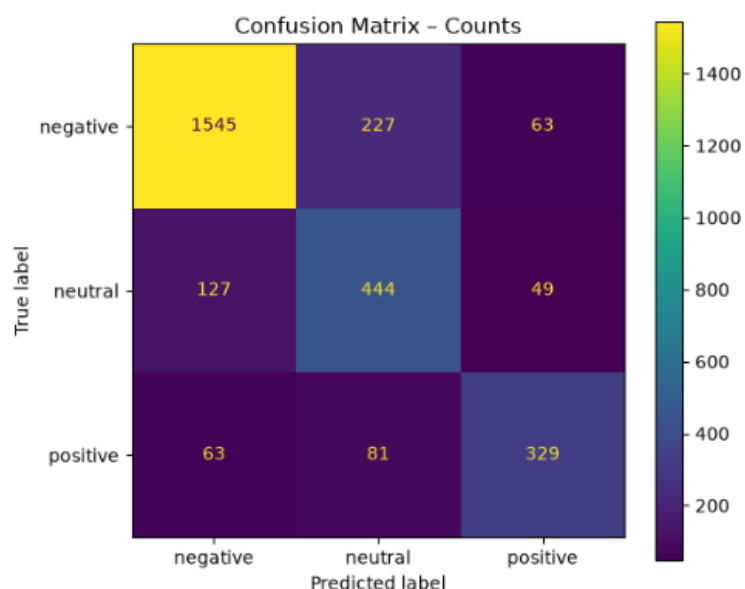


Fig: Count confusion matrix

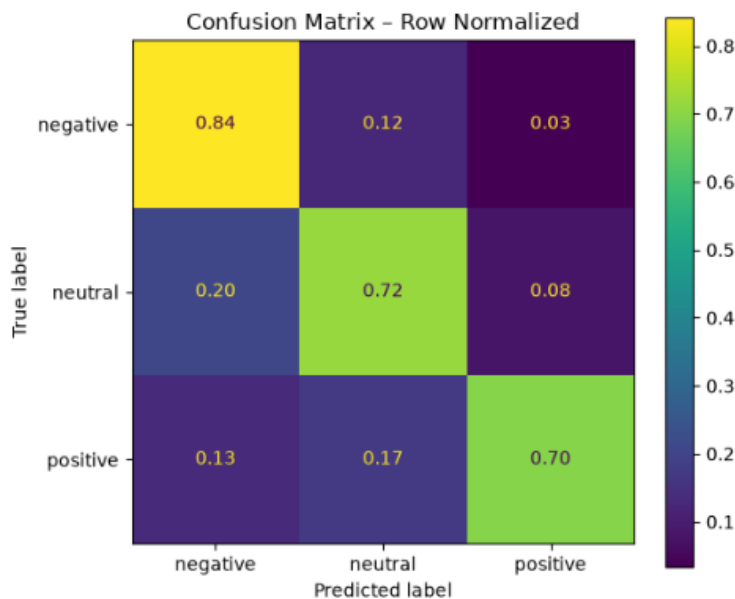


Fig: Row-normalized confusion matrix

The final test set contains **2,928 observations**, of which **2,318 are correctly classified** and **610 are incorrectly classified**.

Table 10.1 Final Error Summary

Error-analysis item	Verified value
Correct predictions	2,318
Incorrect predictions	610
Test observations	2,928
Test error rate	0.2083

Thus, the classifier has an overall test error rate of approximately **20.83%**.

The class-wise results indicate that the main difficulty is associated with distinguishing **neutral sentiment** from other sentiment categories. This is expected because neutral tweets may contain words that are also frequently associated with positive or negative airline experiences. Informal wording and limited context can make these distinctions more difficult.

The notebook also performs representative error inspection. **Ten incorrect predictions** were sampled using `random_state=42` and saved as `inspected_errors_10.csv`. The inspected examples include cases where:

- positive tweets were predicted as negative,
- neutral tweets were predicted as negative,
- negative tweets were predicted as neutral, and
- negative tweets were predicted as positive.

These errors demonstrate that individual sentiment decisions can be difficult even when the overall classifier performs reasonably well.

For privacy and responsible-use reasons, the report does not reproduce usernames or other potentially identifying information from the tweets. This follows the laboratory requirement that raw personal identifiers should not be exposed in the final report.

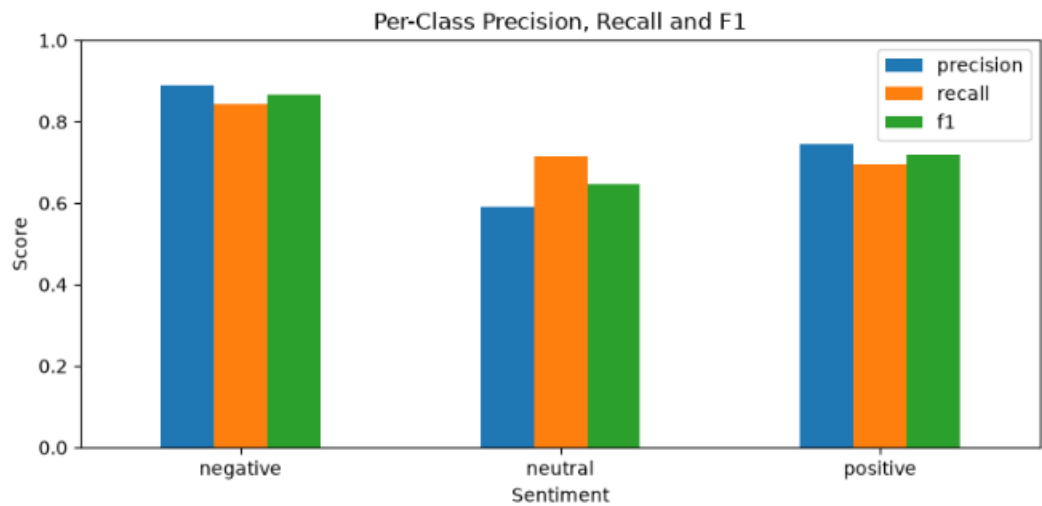
11. Training-Only TF-IDF Top Terms

To improve interpretability, the notebook extracts the highest-scoring TF-IDF terms associated with each sentiment class. Importantly, these terms are calculated using the **training data only**, rather than using the locked test set. This maintains the same leakage-safe principle applied throughout the experiment.

Table 11.1 Top TF-IDF Terms

Class	Top terms reported by notebook
Negative	to, the, flight, and, on, you, for, my, is, in, it, not, of, your, no
Neutral	to, url, you, the, on, can, flight, for, is, it, my, in, do, user user, and
Positive	thanks, thank, thank you, you, user thanks, user thank, the, for, to, great, thanks for, and, it, for the, url

The positive class contains terms such as **“thanks,” “thank you,”** and **“great,”** which are intuitively associated with positive customer experiences. The negative class contains terms such as **“not”** and **“no,”** along with airline-related terms such as **“flight.”** The presence of <URL> and normalized user-related tokens demonstrates that these placeholders remain part of the textual representation after preprocessing.



These terms should not be interpreted as deterministic sentiment rules. They are high-scoring features within the TF-IDF representation and provide an indication of the vocabulary patterns learned by the model.

12. Controlled Preprocessing Ablation

A controlled ablation experiment was conducted to determine whether retaining **hashtags and emojis** contributes to classification performance. The same dataset split and Logistic Regression configuration were maintained, while hashtags and emojis were removed from the text representation.

Table 12.1 Ablation Results

Configuration	Macro F1	Accuracy
Core preprocessing: hashtags/emojis preserved	0.7442	0.7917
Ablation: hashtags/emojis removed	0.7398	0.7879

Removing hashtags and emojis decreases Macro F1 from **0.7442 to 0.7398**, a reduction of **0.0044**. Accuracy decreases from **0.7917 to 0.7879**, a reduction of **0.0038**.

Although the improvement is relatively small, the result supports the decision to retain these social-media-specific features in the core preprocessing pipeline. It also provides experimental evidence for the preprocessing choice rather than relying only on an assumption that emojis and hashtags are useful.

Therefore, the **core preprocessing configuration with hashtags and emojis preserved** is retained for the final reported model.

13. Entity-Specific Performance

Because the dataset contains an airline field, the final model can also be evaluated separately for each airline. This provides an entity-specific view of classification performance and helps identify whether the model behaves consistently across the different service entities represented in the dataset.

Table 13.1 Airline-Level Performance

Airline	N	Accuracy	Error rate
United	774	0.788114	0.211886
US Airways	565	0.824779	0.175221
American	535	0.828037	0.171963
Southwest	506	0.758893	0.241107
Delta	456	0.776316	0.223684
Virgin America	92	0.663043	0.336957

The highest entity-level accuracy is obtained for **American**, with an accuracy of **0.828037**, followed closely by **US Airways** at **0.824779**.

The lowest accuracy is observed for **Virgin America**, at **0.663043**. However, Virgin America has only **92 test observations**, substantially fewer than the other airlines. Therefore, its lower accuracy should be interpreted cautiously because estimates based on smaller samples are less stable.

Southwest has the next-lowest accuracy at **0.758893** and has substantially greater support than Virgin America. This suggests that its errors deserve greater attention in entity-specific analysis.

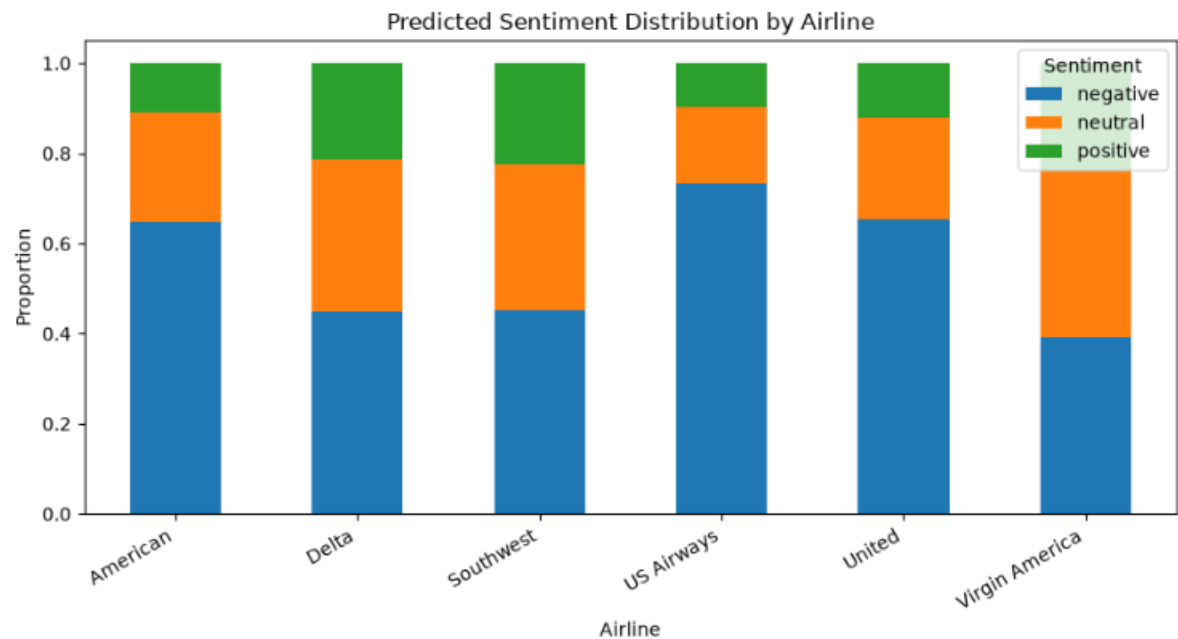
These results should not be interpreted as evidence that the model is inherently better or worse for customers of a particular airline. Differences may result from the number of observations, vocabulary, sentiment distribution, and characteristics of the tweets represented in each entity subset.

13.1 Predicted Sentiment Distribution by Airline

Table 13.2 Predicted Sentiment Proportions

Airline	Negative	Neutral	Positive
American	0.646729	0.242991	0.110280
Delta	0.447368	0.339912	0.212719
Southwest	0.450593	0.324111	0.225296
US Airways	0.734513	0.168142	0.097345
United	0.653747	0.224806	0.121447
Virgin America	0.391304	0.369565	0.239130

The predicted distributions vary considerably across airlines. For example, **US Airways** has the highest predicted negative proportion at **0.734513**, while **Virgin America** has the lowest negative proportion at **0.391304**.



However, these percentages represent the **model's predictions within this dataset**, not the actual sentiment of all airline customers. Social-media samples may be affected by selection bias and therefore should not be interpreted as market-share or customer-satisfaction estimates.

14. Linguistic Challenge Analysis

Table 14.1 Challenge Analysis

Challenge type	N	Accuracy	Error rate
Negation	615	0.874797	0.125203
Sarcasm hint	3	0.333333	0.666667
Emoji-heavy	108	0.666667	0.333333
Hashtag	452	0.780973	0.219027
Product/brand handle	2,928	0.791667	0.208333
Elongation	330	0.781818	0.218182
Mixed sentiment hint	6	0.833333	0.166667

Tweets containing **negation** achieve relatively strong accuracy of **0.874797**, suggesting that the TF-IDF model can capture some common negation-related patterns.

Emoji-heavy tweets are more difficult, with an accuracy of **0.666667** and an error rate of **0.333333**. This suggests that emoji-related sentiment may not always be adequately represented by conventional word-level TF-IDF features.

The sarcasm-hint subset has an accuracy of only **0.333333**. However, only **three tweets** satisfy this heuristic, so this result cannot support a general conclusion about the model's ability to recognize sarcasm. Similarly, the mixed-sentiment subset contains only **six observations** and should not be generalized beyond this dataset.

The product/brand-handle category contains **2,928 observations**, meaning that the heuristic used in the notebook flags the entire test set. Consequently, its accuracy is identical to the overall test accuracy and should not be interpreted as evidence of a distinct difficult subset.

15. Confidence Analysis

The selected Logistic Regression model provides class probabilities, allowing the notebook to examine the relationship between prediction confidence and classification accuracy.

The **mean maximum predicted probability is 0.6948**. The test observations are divided into five confidence groups using probability quintiles.

Table 15.1 Confidence Analysis

Confidence quintile	N	Accuracy
(0.343, 0.526]	586	0.535836
(0.526, 0.638]	585	0.721368
(0.638, 0.754]	586	0.805461
(0.754, 0.871]	585	0.928205
(0.871, 0.998]	586	0.967577

A clear relationship can be observed between model confidence and accuracy. The lowest-confidence group achieves only **0.535836 accuracy**, whereas the highest-confidence group achieves **0.967577 accuracy**.

This indicates that the model's confidence ranking is useful for identifying predictions that are more likely to be correct. However, the analysis does **not establish that the probabilities are calibrated**. Formal calibration techniques such as reliability diagrams or calibration-error measurements were not performed in the notebook.

16. Reproducibility and Computational Issues

Reproducibility is an important requirement of the experiment. The notebook uses a fixed random seed and stores the selected model pipeline so that the trained model can be reused and checked.

Table 16.1 Reproducibility Configuration

Item	Verified notebook result
Random seed	42
Core split	80:20 stratified
CV	5-fold StratifiedKFold
Selected model	Logistic Regression
Model artifact	selected_pipeline.joblib
Reload verification	PASSED
Output directory	lab05_outputs

The notebook successfully saves the selected pipeline and reloads it to verify that predictions produced by the reloaded model match the predictions from the original model. This provides a direct check that the saved artifact preserves the trained model's behavior.

During notebook execution, two resource-related issues were encountered. An earlier five-fold cross-validation attempt using `n_jobs=-1` produced a **MemoryError/BrokenProcessPool**, indicating that parallel processing exceeded the available memory. A separate matplotlib figure-saving operation also produced a **MemoryError** during PNG rendering.

These errors are computational/environmental issues and should not be confused with model performance errors. The successful CV results and model outputs reported in this document come from the completed notebook execution rather than from the failed parallel attempt.

17. Advanced A – BiLSTM

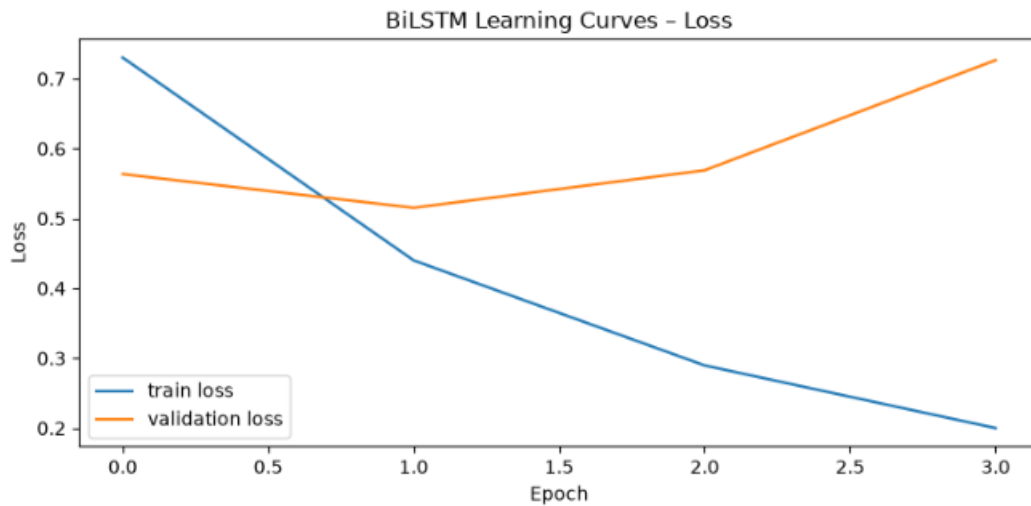
A **Bidirectional Long Short-Term Memory (BiLSTM)** model was implemented as an advanced extension to determine whether a sequence-based neural architecture could improve upon the classical TF-IDF models.

The model uses a vocabulary limit of **30,000 words**, a maximum sequence length of **80**, and an embedding dimension of **128**. The architecture includes a **Bidirectional LSTM with 64 units**, dropout, dense layers, and a softmax output layer for the three sentiment classes. Training uses early stopping and model checkpointing.

Table 17.1 BiLSTM Results

Metric	Verified value
Training epochs completed	4 of maximum 10
Training time	146.72 seconds
Test Macro F1	0.7378
Test Weighted F1	0.7910
Test Accuracy	0.7883
Batch inference time	5.8174 seconds

The BiLSTM achieved a Macro F1 of **0.7378**, which is lower than the Logistic Regression Macro F1 of **0.7442**. Its accuracy of **0.7883** is also slightly below Logistic Regression's **0.7917**.



Therefore, the completed experiment does **not provide evidence that the BiLSTM improves upon the selected classical model**. This result is important because it demonstrates that increasing model complexity does not automatically result in improved performance, particularly when a strong TF-IDF representation and linear classifier are already effective for the dataset.

18. Advanced B – Twitter-domain Transformer

The advanced Transformer experiment uses **BERTweet (vinai/bertweet-base)**, a pretrained Transformer checkpoint designed for Twitter-domain language. The objective was to investigate whether contextual subword representations could provide an improvement over both classical TF-IDF models and the BiLSTM.

Table 18.1 Transformer Experiment Status

Component	Verified notebook status
Checkpoint	vinai/bertweet-base
Label mapping	0=negative, 1=neutral, 2=positive
Transformer training set	9,955
Validation set	1,757
Locked test set	2,928
Maximum sequence length	128
Tokenizer	Completed
TrainingArguments / Trainer training	Failed before training
Reason	NameError: TrainingArguments is not defined
Locked-test Transformer metrics	Not available

The Transformer preprocessing stage was successfully prepared. The locked test set remained unchanged, while the training data was further divided into training and validation subsets.

However, the Transformer training cell did not execute because TrainingArguments was not defined in the notebook's executed state. The subsequent evaluation could not proceed because the trainer object was therefore not created.

Consequently, **no Transformer Accuracy, Macro F1, or Weighted F1 is reported**. Reporting a numerical Transformer result without successful training and evaluation would not be supported by the notebook and would compromise the reproducibility of the report.

The notebook environment recorded **TensorFlow 2.21.0, Transformers 5.14.1, Datasets 5.0.1, and Evaluate 0.4.6**. During tokenization, an emoji-package warning was also produced, although tokenization itself completed.

19. Advanced Uncertainty Requirement

The advanced experiment includes a framework for evaluating neural models across multiple random seeds. Such an analysis is useful because neural-network performance can vary depending on initialization and training randomness.

However, although the notebook contains the structure for a **three-seed evaluation**, the actual seed-43 and seed-44 experiments were not completed. Therefore, a three-seed mean and standard deviation cannot be calculated from the available executed results.

Accordingly, this report reports the **completed BiLSTM result only** and does not present an unsupported uncertainty estimate for the advanced models.

20. Limitations and Responsible Use

Several limitations should be considered when interpreting the results.

- The dataset represents tweets collected in **2015**, and therefore its language and user behavior may differ from contemporary social-media communication.
- The dataset is substantially **imbalanced**, with negative tweets representing 62.69% of all observations.
- The audit identified **155 duplicate tweet IDs and 213 duplicate raw-text records**. These were documented but not reported as removed because the notebook does not establish that they were removed.
- Sentiment labels may contain **annotation ambiguity or noise**.
- Sarcasm and mixed sentiment are difficult linguistic phenomena, and the corresponding challenge subsets in this experiment have very small sample sizes.
- Entity-level differences may reflect differences in sample composition rather than genuine differences in customer sentiment.
- The confidence analysis demonstrates a useful relationship between confidence and accuracy but does not establish probability calibration.
- Performance may change when the model is applied to newer tweets, different platforms, different domains, or substantially different linguistic styles.
- Social-media sentiment proportions should not be interpreted as representative estimates of the opinions of all airline customers.

- The classifier is intended for **aggregate analytical and educational use**, rather than autonomous decisions or actions against individual users.

These limitations are consistent with the laboratory's emphasis on sampling bias, domain shift, temporal validity, annotation quality, and responsible interpretation of social-media sentiment.

21. Model Card

Table 21.1 Model Card

Item	Verified description
Task	Three-class airline tweet sentiment classification
Dataset	Twitter US Airline Sentiment
Classes	Negative / Neutral / Positive
Core selected model	Logistic Regression
Representation	TF-IDF word 1–2 grams
Selection metric	5-fold CV Macro F1
CV Macro F1	0.744384 ± 0.008241
Locked-test Macro F1	0.7442
Locked-test Accuracy	0.7917
Locked-test Weighted F1	0.7958
Advanced BiLSTM Macro F1	0.7378
Transformer	Prepared but not trained successfully
Intended use	Aggregate sentiment analysis / educational research
Out-of-scope	Autonomous action against individuals

The model card summarizes the final system and its intended use. The Logistic Regression model is preferred for the completed experiment because it combines strong performance, relatively low computational requirements, reproducibility, and interpretability through its TF-IDF representation.

22. Conclusion

This experiment developed and evaluated a reproducible three-class sentiment-classification system for airline-related tweets. The complete workflow included dataset auditing, minimal tweet preprocessing, leakage prevention, stratified train-test splitting, five-fold cross-validation, baseline comparison, classical machine-learning classification, final locked-test evaluation, entity-level analysis, linguistic challenge analysis, confidence analysis, and advanced neural-model experimentation.

Among the classical learned models, **Logistic Regression was selected as the final model** because it achieved the highest five-fold cross-validation Macro F1 of **0.744384 ± 0.008241**. On the locked test set of **2,928 tweets**, it achieved an **Accuracy of 0.7917, Macro F1 of 0.7442, and Weighted F1 of 0.7958**.

The class-wise results demonstrate that the model performs particularly well on **negative sentiment**, with an F1-score of **0.865546**, while **neutral sentiment** remains the most challenging class, with an F1-score of **0.647230**. This confirms the importance of using Macro F1 and class-specific evaluation rather than relying solely on overall accuracy.

The controlled preprocessing experiment also demonstrated that retaining hashtags and emojis provides a small performance benefit. Macro F1 decreased from **0.7442 to 0.7398** when these features were removed. Entity-level analysis showed differences in performance across airlines, while the linguistic challenge analysis identified emoji-heavy and sarcasm-hint tweets as potentially difficult cases. However, the very small sarcasm and mixed-sentiment subsets prevent strong generalization from those results.

The **BiLSTM extension achieved a Macro F1 of 0.7378**, which was slightly lower than Logistic Regression. Thus, the more complex sequence model did not improve performance in the completed experiment. The BERTweet Transformer extension was prepared but could not be trained because TrainingArguments was undefined. Therefore, no Transformer performance is claimed.

Overall, the experiment demonstrates that a carefully constructed **TF-IDF + Logistic Regression pipeline provides a strong, efficient, and reproducible solution** for this dataset. More importantly, the study demonstrates that reliable NLP experimentation requires not only high predictive performance but also **proper data auditing, leakage control, consistent validation, class-aware metrics, error analysis, reproducibility checks, and responsible interpretation of social-media data**.

GitHub: https://github.com/Shobana0212/lab5_Tweet-Sentiment-Analysis.git